Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/19/2023
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	RNA-Seq Guided Mutant Analysis to Discover New Components of Gravity Signaling in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	03/01/2021	**End Date:**	02/29/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	15	**Monitoring Center:**	NASA KSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Genes that show contrasting expression across the gravity continuum or between wild-type and gravity perception mutants represent candidates for discovery of novel components of the gravity perception and response pathway in plants. We hypothesize that these genes will reveal novel contributors to gravity perception, signal transduction, and response regulation. We have already completed the necessary RNA-Seq-based contrast analysis and have genes lists in hand to be further studied. The roles and contributions of these genes will be systematically tested in the project proposed here through a series of physiological, cell biological, and molecular screens using knockout mutants. Each mutant will be systematically tested for a gravity-related phenotype using high-resolution image analysis approaches developed in the Principal Investigator's lab. The role of the gene product in controlling differential auxin transport will be assessed by introducing an auxin reporter gene into each mutant background and quantifying expression of the reporter using confocal microscopy. The results of these screens will allow for the mapping of each gene product onto the gravity response pathway, leading to significant insights into plant gravity signaling in support of the development of gravity-altered plants tuned for future spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although gravity is a constant stimulus for plant growth and development, we do not have a full account of how plants detect this stimulus and convert it into molecular, cellular, and physiological information. This research seeks to greatly expand our understanding of how plants sense and respond to this fundamental force. An enhanced understanding of the mechanisms employed by plants to sense and respond to gravity will lead to a more complete and successful set of criteria for engineering crop plants for Earth and future missions off-planet.
Task Progress:	In this year of funding (March 2023 - March 2024), we focused on screening another 20 candidate genes from our RNA-seq contrast list using T-DNA insertional mutants from the Salk collection. In previous work, we screened approximately 70 T-DNA lines for mutant phenotypes, finding 11 mutant lines with putative gravity-involved phenotypes for further analysis. To extend this work, we received seeds from the Arabidopsis Biological Resource Center (ABRC) for an additional 20 lines in summer 2023. The work was carried out by 5 undergraduate research students, two of whom had prior research experience. Students performed all aspects of the project, including preparing sterile growth media and sowing seeds, imaging experiment setup and image analysis, DNA and RNA isolations, and polymerase chain reaction (PCR) or reverse transcription-polymerase chain reaction (RT-PCR) experiments. Thus far, we have 88 transfer DNA (T-DNA) mutant lines on hand and have successfully completed genotyping 65. Of the lines screened thus far, 5 have been confirmed heterozygous for the T-DNA allele, and of those, 2 show a gravity related phenotype; 48 lines have been confirmed homozygous for the T-DNA carrying allele; the remainder of the lines genotyped have shown ambiguous results and we are in the process of redesigning primers. Some lines without a positive confirmation PCR result may be due to a later transposition event with the T-DNA. We are in the process of amplifying and sequencing the region surrounding the location of the T-DNA insertion to detect possible footprint deletions. Some of the lines also showed a lack of positive product for the positive control primer pair as well, indicating bad primer design or PCR conditions. Sequencing in the region of the purported T-DNA will also shed light on these lines. In addition to genotyping, we are also testing for transcript knock-down in the T-DNA lines to be able to correlate any phenotypes observed with lack of transcript abundance. To date, we have carried out RT-PCR on 35 lines and confirmed the lack of mRNA transcript for the gene of interest in 13 of our lines, with 15 lines showing no difference in transcript abundance compared to the wild-type. The RT-PCR experiments required for this assay are more demanding than the genotyping PCR experiments, leading to an overall slower rate of completion. We carried out this assay, along with many others, as part of my Plant Physiology lab experience for 16 students in Spring 2023. I have redesigned the lab experience around this NASA-funded screening project to provide students hands-on experience with real research. Since so many of the assays and screens of these mutant lines is routine in the lab, this will provide students the opportunity to contribute to the project while learning valuable skills in bioinformatics, DNA and RNA isolation, real-time RT-PCR, and various plant growth analyses. I incorporated this project into the lab portion of the class last spring and it was moderately successful, with about half of the data generated by student groups of sufficient quality to be useful. This spring I will be structuring the work a bit more and providing templates for the students to collect and interpret results, which will hopefully mitigate some of the loss I saw last year. In addition to the molecular characterization of mutants, we have made good progress on the growth experiments designed to test the mutants for any gravity-related phenotypes. We have developed a novel assay to detect differences in the regulation of differential growth. The assay is based on a previous technique used heavily in my lab, called ROTATO, which uses real-time image analysis to constrain the root tip at a constant angle over time by rotating the plate. This technique is good for measuring single roots, but does not scale well to large numbers of roots on a plate, as required for a medium-scale mutant screening project. I have used the average response rate (rotation) of wild-type roots as an input to control rotation of whole plates of mutant seedlings, reasoning that if the roots have “normal” gravity response, their tip angles will remain relatively unchanged because they are able to maintain the differential growth required to do so. We have now completed analysis of 50 T-DNA mutant lines using this technique and have identified 18 lines that show a statistically significant difference in response compared to wild-type; 13 of the 18 lines show significantly reduced gravity response, while the other 5 lines show greater gravity response. Other experiments to characterize the mutants include standard reorientation assays to assess gravitropic response and root phototropic assays to try to distinguish gravity perception processes from growth regulation and gravity response elements. We have conducted hundreds of standard root reorientation assays thus far, and we are in the process of measuring tip angles and collating results. For the root phototropism assays, we have done some pilot testing with large-scale experiments, but I am unsatisfied so far with the amount of curvature induced in the positive and negative controls. Last summer, one of the student researchers worked on developing a better assay design and construction that will optimize this response, but we have not yet screened a large number of the mutants yet. This year we designed and cloned Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) constructs to knock out genes for which there is no T-DNA line available in a stock center. Experiments to transform these constructs into plants failed over the summer when the candidate plants were left untended over a weekend and there was miscommunication about who was scheduled to water, unfortunately. On the positive side, I was pleased with how straightforward the cloning process was to create the constructs and remain positive that this will allow us to extend the screen to genes with no knockout mutants available. This could also allow us to generate second alleles for those genes for which we find a phenotype and wish to confirm. I will be returning to this knockout strategy with my Plant

	Physiology lab students in Spring 2024 to allow for progress on this objective. In addition to using CRISPR to knock out genes of interest, we have begun making double-mutants between those mutants that show a promising gravity-related phenotype. By testing the phenotypes of double mutants in the xxx mutant background, we hope to identify molecular participants in the gravity sensing pathway by finding double mutants with a stronger phenotype than either single mutant alone. We have crossed 23 of our confirmed mutants into the mutant to date, with 18 confirmed F1 lines. In addition, we are in the process of screening 4 F3 lines collected from individual F2 parents to identify double mutants. Rather than attempt to bring in a postdoc to fill this position, I was able to hire one of our existing part-time department lab managers for the rest of her work hours. This has worked out well thus far, as she is present every day of the week and can assist with planting for experiments and plant care tasks. I am beginning to train her on DNA and RNA isolation protocols and I look forward to her being able to assist with these tasks in the future.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Barker R, Kruse CPS, Johnson C, Saravia-Butler A, Fogle H, Chang HS, Trane RM, Kinscherf N, Villacampa A, Manzano A, Herranz R, Davin LB, Lewis NG, Perera I, Wolverton C, Gupta P, Jaiswal P, Reinsch SS, Wyatt S, Gilroy S. "Meta-analysis of the space flight and microgravity response of the Arabidopsis plant transcriptome." npj Microgravity. 2023 Mar 20;9(1):21. https://doi.org/10.1038/s41526-023-00247-6 ; PMID: 36941263; PMCID: PMC10027818 , Mar-2023
Articles in Peer-reviewed Journals	Roychoudhry S, Sageman-Furnas K, Wolverton C, Grones P, Tan S, Molnár G, De Angelis M, Goodman HL, Capstaff N, Lloyd JPB, Mullen J, Hangarter R, Friml J, Kepinski S. "Antigravitropic PIN polarization maintains non-vertical growth in lateral roots." Nat Plants. 2023 Sep 4. https://pubmed.ncbi.nlm.nih.gov/37666965 ; PMID: 37666965; PMCID: PMC10505559 , Sep-2023